

LeetCode 1725

Number Of Rectangles That Can Form The Largest Square

Approach & Logic

Two-Pass Strategy:

- **First Pass:** Determine the maximum possible side length (`maxlen`) of a square that can be extracted from any given rectangle. For any rectangle `[length, width]`, the largest square we can form has a side length of `min(length, width)`.
- **Second Pass:** Iterate through the rectangles again to count how many of them can form a square of the identified `maxlen`.

C++ Source Code

```
class Solution {
public:
    int countGoodRectangles(vector<vector<int>>& rectangles) {
        int maxlen=INT_MIN;

        for(auto it:rectangles){

            int a=it[0];
            int b=it[1];

            maxlen=max(maxlen,min(a,b)); //ek [2,99] ke rectangle
            se 2 side length ka square bana skte he
        }

        int ans=0;
        for(auto it:rectangles){

            int a=it[0];
            int b=it[1];
            if(min(a,b)==maxlen) ans++;
        }

        return ans;
    }
};
```